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LOGISTICS, DELIVERY & RETURNS INTELLIGENCE

Executive Performance & Data Reliability Review

Verified Python analysis of fulfillment status, recorded delivery timing, courier comparisons, return requests, and underlying data reliability.

PREPARED FOR Business Ownership & Operations Leadership

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Illustrative logistics image; all numeric findings derive from the supplied datasets.

02 / CONTENTS

Contents & Navigation

A decision-first report with detailed technical evidence and traceable results. Select a section below to jump to it.

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Reading guide: Pages 3–5 summarize the operational story. Pages 6–12 explain data preparation and reliability. Pages 13–17 show the five required charts. Pages 18–25 connect findings to decisions and reproducibility.

03 / EXECUTIVE SUMMARY

Executive Summary

01 The assessment covers seven linked CSV sources and 15,000 unique orders.

02 All 15,000 prepared shipment records are included in shipment-status reporting.

03 11,680 shipments (77.87%) are recorded as Delivered; this is a status share, not an on-time rate.

04 2,694 shipments (17.96%) are In Transit; 161 (1.07%) are Lost.

05 The baseline usable delivery-time sample contains 11,122 records with a 4.00-day mean.

06 After chronology screening, 5,667 records remain with a 4.02-day mean.

07 The lowest observed courier mean shifts after screening; differences are small.

08 1,200 orders (8.00%) have a recorded return request; this is not a completed-return rate.

09 Changed mind is the leading recorded reason: 182 requests (15.17% of the 1,200).

10 Home & Kitchen has an 8.64% order-category request association, not an item-level return rate.

11 Cross-system event-date conflicts warrant reconciliation before operational benchmarking.

Executive decision: Prioritize source chronology/status reconciliation and returned-item identification before using these descriptive results to compare service providers or prescribe product-category interventions.

04 / PROJECT OVERVIEW

Business Questions & Scope

Business goal: Answer “How well are we delivering, and what causes returns?” using linked order, shipment, product, payment, and recorded return-request data.

SHIPMENT EXECUTION

What do recorded Delivered, In Transit, Lost and Returned to Sender states reveal about fulfillment visibility?

DELIVERY-TIME RELIABILITY

How many records have usable date pairs, and how sensitive are averages to order–shipping chronology checks?

COURIER COMPARISON

How do the five couriers compare on the eligible sample, without overclaiming small differences?

RETURN INTELLIGENCE

Which recorded reasons are most common, and what do order-category associations reveal or fail to reveal?

Analytical boundary: The datasets do not contain verified delivery promises/SLA attainment, confirmed returned-item IDs, or post-intervention cost savings. The report does not infer those outcomes.

05 / EXECUTIVE KPI DASHBOARD

Executive Operating Snapshot

ORDERS	DELIVERED	DELIVERY MEAN	RETURN REQUESTS
15,000	77.87%	4.00 days	8.00%
Distinct order IDs	11,680 recorded shipments	11,122 eligible records	1,200 distinct orders

Recorded operational metric	Verified total	Interpretation
In Transit	2,694 / 15,000 = 17.96%	Current recorded shipment state; not automatically late.
Lost	161 / 15,000 = 1.07%	Recorded status; source consistency needs validation.
Returned to Sender	465 / 15,000 = 3.10%	Shipment outcome; separate from customer return requests.
Chronology-screened lead time	n = 5,667; mean = 4.02 days	Alternative analytical subset, not an after-period improvement.
Top return reason	Changed mind: 182 requests	15.17% of recorded requests; reasons are not verified causes.

Interpretation: On-time delivery rate cannot be calculated without a validated promised delivery date and reliable actual-delivery evidence. “Delivered” is a recorded status, not a promise-attainment measure.

06 / DATA LINEAGE

Data Sources & Reconciliation

Original CSV files were loaded and prepared in Python. Exact-row duplicates were removed from working copies, while unresolved source exceptions were retained/flagged. Source files were not altered.

Source dataset	Original rows	Exact duplicates removed	Prepared rows
Customers	5,050	50	5,000
Products	800	0	800
Orders	15,075	75	15,000
Order Items	37,763	262	37,501
Payments	15,000	0	15,000
Shipments	15,000	0	15,000
Returns	1,200	0	1,200
TOTAL	89,888	387	89,501

Source lineage: 89,888 input rows − 387 exact duplicate rows = 89,501 prepared rows across the seven different data entities. This is not an order count, because the tables represent different relationships.

07 / CLEANING METHODOLOGY

Six Mandatory Cleaning Controls

The six required rules are documented in the actual Python notebook and validated in the Step 108 evidence exports. Each rule below distinguishes a data correction from an unresolved exception.

01 Exact duplicate removal

Use all original columns to identify duplicates; keep first; 387 removed.

02 Missing-value treatment

Profile 13 fields containing 6,345 missing cells; retain, flag, or add separate reporting labels based on data use.

03 Product category normalization

Retain source category and create canonical category_clean; 137 source labels changed.

04 Mixed-format date parsing

Create typed date columns for order, shipping, delivery and payment fields used in the Option B analysis.

05 Currency-to-number parsing

Parse purchase unit-price strings into numeric values without inventing missing prices.

06 Join-key and orphan checks

Validate six directional relationships; retain/flag unmatched customer and product references.

Scope: “Cleaned” means prepared with documented transformations and quality flags. It does not mean the source records passed every business consistency check.

08 / DATA PREPARATION

Duplicate Removal & Category Mapping

Deduplication applies an exact-row definition, not a broad “same customer or order ID” deletion rule. Standardized product categories are added as a new reporting field.

Source group	Raw rows	Duplicates removed	Prepared rows
Customers	5,050	50	5,000
Orders	15,075	75	15,000
Order items	37,763	262	37,501
Remaining four datasets	32,000	0	32,000

CATEGORY STANDARDIZATION / ORIGINAL → CANONICAL

Source label → Reporting label	Changed	Source label → Reporting label	Changed
Book → Books	20	Beauty & Care → Beauty	11
electronics → Electronics	20	Toy → Toys	11
Home and Kitchen → Home & Kitchen	19	Elec → Electronics	11
Electronics [trailing space] → Electronics	17	Fashon → Fashion	8
Grocery → Groceries	12	Sports & Outdoor → Sports	8

Result: Eight standardized reporting categories. Product ID mappings that cannot be joined are separately labeled Unknown / Unmapped rather than assumed to belong to any category.

09 / DATA PREPARATION

Missing-Value Decisions

A missing source value should not be silently transformed into zero. Decisions below preserve analytical meaning and document what fields were or were not used.

Dataset	Source field	Missing	Documented decision
Customers	gender	298	Retain missing; not used for Option B KPI
Customers	age	149	Retain missing; not used for Option B KPI
Customers	email	104	Retain missing; not used for Option B KPI
Customers	phone	145	Retain missing; not used for Option B KPI
Customers	loyalty_tier	264	Retain original; add separate Unknown label
Products	stock_qty	19	Retain missing; not used for Option B KPI
Orders	discount_pct	756	Retained; missing-discount flag
Orders	shipping_fee	470	Retained; missing-fee flag
Order Items	unit_price_at_purchase	2,236	Retain; exclude from price calculations
Payments	amount_paid	878	Retain missing; not used for Option B KPI
Shipments	delivered_date	729	Retain and flag; exclude from lead time
Returns	refunded_flag	110	Retain; add Unknown and review flag
Returns	refund_amount	187	Retain; refund review/amount flags

Reconciled total: 13 source fields contain 6,345 missing values after deduplication. Retained nulls are visible analytical limitations rather than quietly fabricated observations.

10 / DATA PREPARATION

Date & Monetary Parsing

Parsing turns source date or purchase-price strings into typed analytical values. Parsing does not automatically repair a logically impossible recorded date.

Date field	Original column	Python treatment	Nonblank parse failures
Order created date	orders.order_datetime	Parsed for joins / chronology	0 additional parse failures
Ship date	shipments.ship_date	Parsed for lead-time computation	0 additional parse failures
Delivered date	shipments.delivered_date	Parsed when present	0 additional parse failures
Payment date	payments.payment_date	Parsed for event sequencing	0 additional parse failures

Missing vs. parsing failure: 729 source delivery dates were already blank. Zero new parse failures among nonblank values does not mean every shipment has a valid delivered date.

PURCHASE-PRICE STRING CONVERSION

Python created unit_price_numeric from order_items.unit_price_at_purchase by removing currency presentation characters and parsing numeric values. 2,236 already missing purchase unit-prices remained missing; no zero-value substitution was introduced. The price conversion checks passed for nonmissing source strings.

Field scope: Some other monetary/date columns not required for the logistics/returns KPIs remain in their source representations. The report does not claim universal standardization.

11 / DATA PREPARATION

Join Integrity & Orphan Records

An orphan is a row whose referenced parent key does not exist in the linked table. The analysis retains flagged rows rather than silently removing them.

Child → parent relationship	Rows checked	Orphan records	Handling
Order → Customer	15,000	43	Preserve order; mark unmatched customer
Order Item → Product	37,501	129	Preserve item; category may be Unknown / Unmapped
Order Item → Order	37,501	0	No unmatched order reference
Payment → Order	15,000	0	No unmatched order reference
Shipment → Order	15,000	0	No unmatched order reference
Return Request → Order	1,200	0	No unmatched order reference

Decision: Do not interpret missing join keys as a reason to fabricate customer/product identities. Join integrity is tracked separately from delivery-performance and return metrics.

12 / DATA RELIABILITY

Source-Data Quality Exceptions

These checks reveal cross-table and timestamp inconsistencies requiring review. A record can appear under several indicators; percentages must not be summed into unique affected orders.

Source quality indicator	Flagged	Checked	Flagged share
Payment date before order	7,436	15,000	49.57%
Shipping date before order	7,397	15,000	49.31%
Shipment status/date conflict	3,707	15,000	24.71%
Cancelled order with Delivered shipment	938	15,000	6.25%
Cancelled, Delivered and Paid	782	15,000	5.21%

PAYMENT BEFORE ORDER

49.57%

7,436 source records

SHIPPING BEFORE ORDER

49.31%

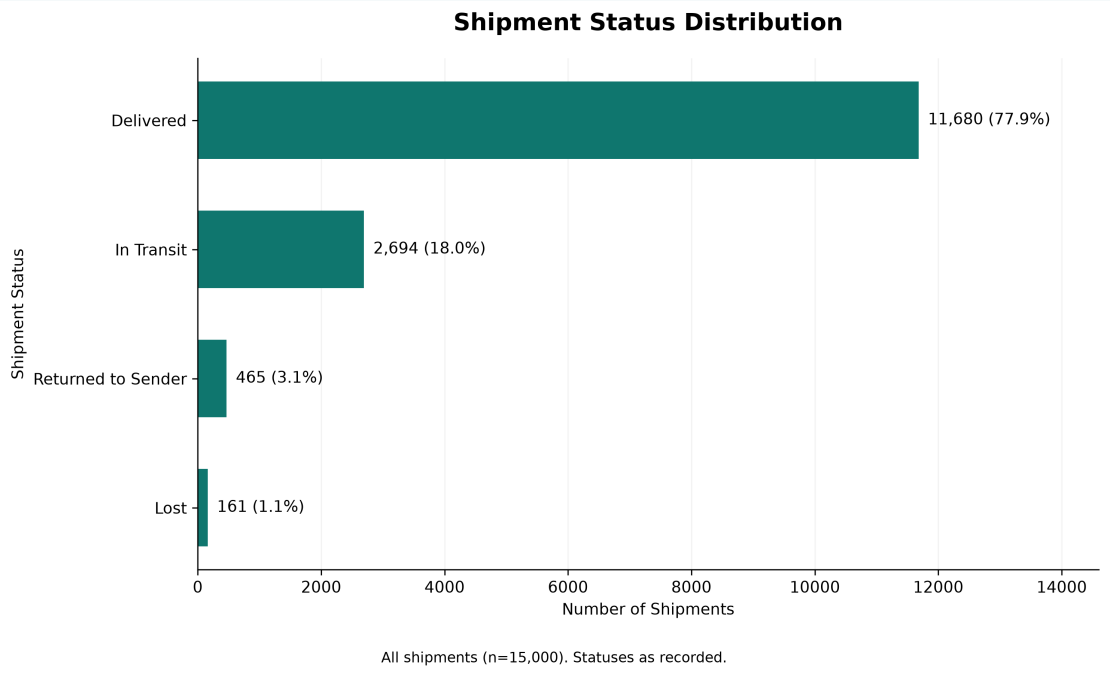
7,397 source records

Quality conclusion: Review event-time conventions, import sequencing, and status transitions before declaring on-time performance or material carrier differences. The flags identify review candidates, not proven business errors.

13 / SHIPMENT ANALYTICS

Shipment Status Breakdown

All 15,000 shipment records are included in this status-distribution chart.



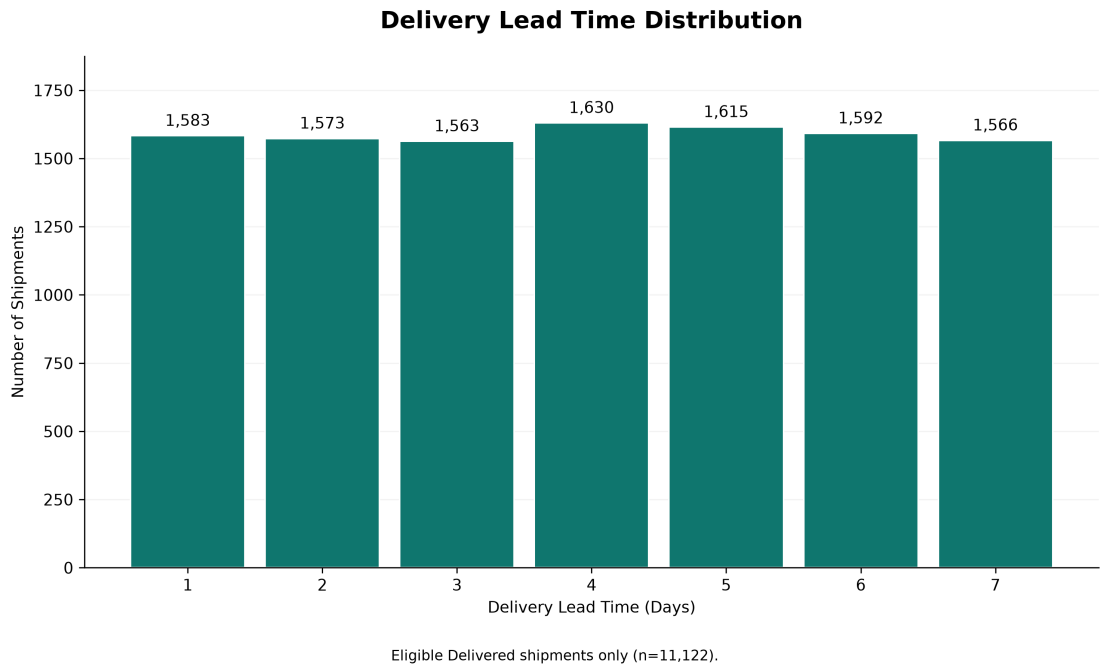
Business interpretation: Delivered dominates the recorded mix: 11,680 / 15,000 = 77.87%; In Transit 17.96%, Returned to Sender 3.10%, and Lost 1.07%.

Reporting limit: A Delivered status is not synonymous with a verified delivery date or a proven on-time delivery outcome.

14 / DELIVERY ANALYTICS

Recorded Delivery Lead-Time Distribution

Recorded days between shipping and delivery, restricted to 11,122 eligible records with usable date pairs.



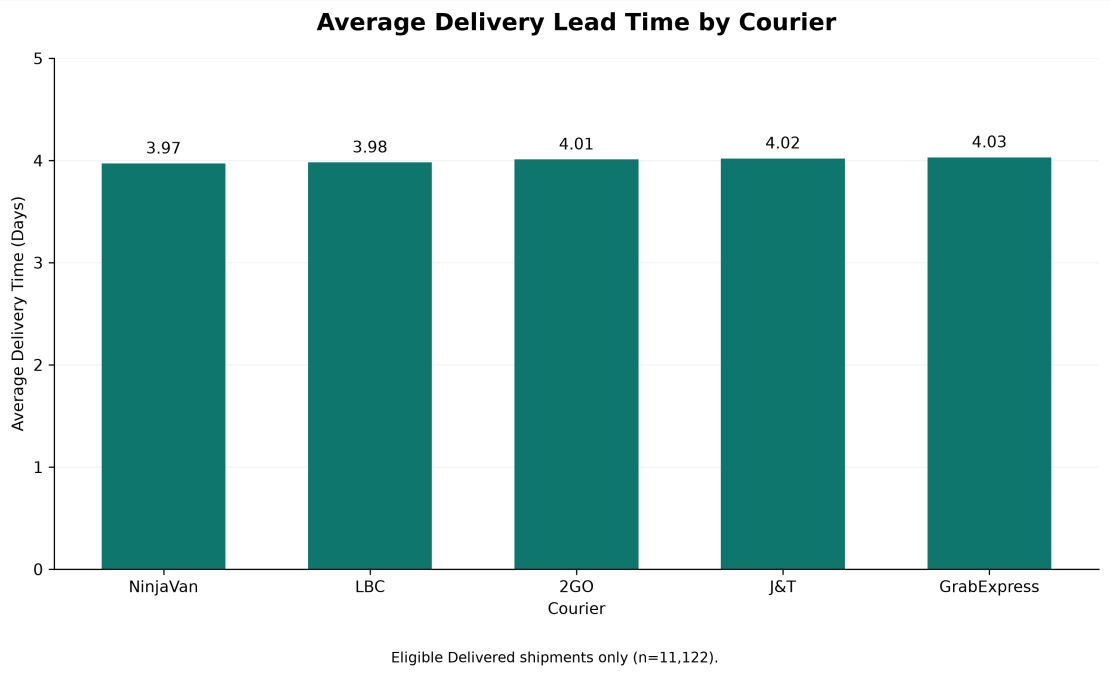
Business interpretation: Mean = 4.00 days; median = 4 days. The chart communicates the distribution of the eligible analytical subset.

Reporting limit: The baseline sample excludes shipments without adequate dates or other eligibility; the dataset still has unresolved chronology conflicts.

15 / COURIER ANALYTICS

Courier Performance Comparison

Eligible delivery lead times grouped by courier, shown as descriptive averages.



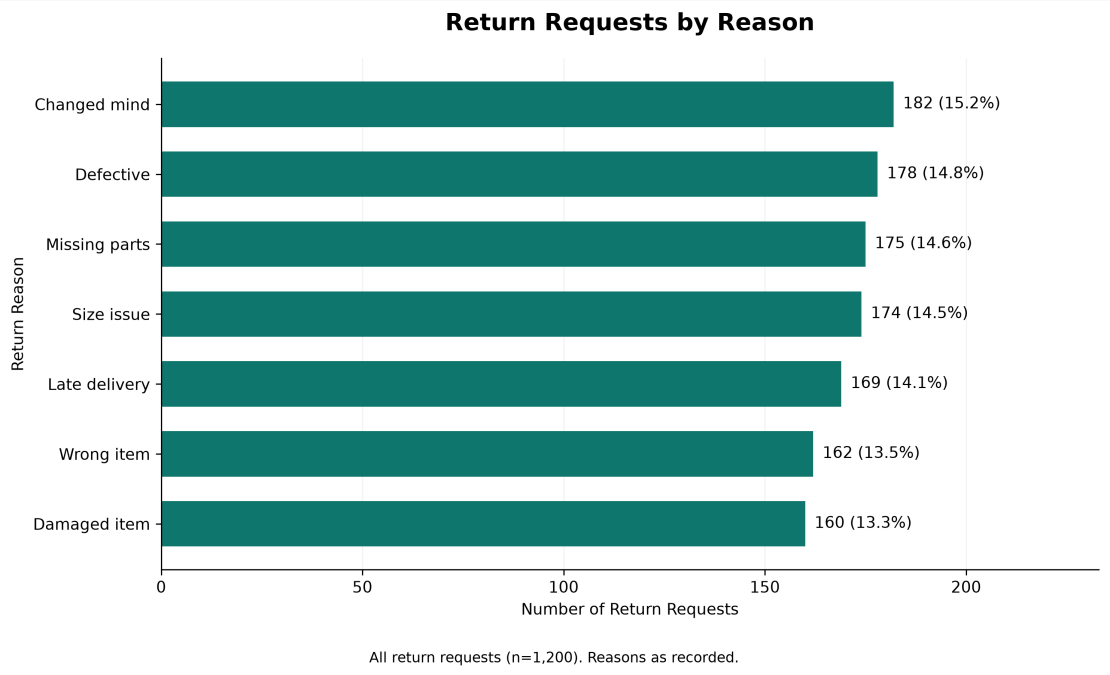
Business interpretation: The lowest observed baseline mean is NinjaVan; after chronology-screening the lowest is LBC. Rankings are sensitive to data-screening rules.

Reporting limit: Differences are small; this chart does not establish a statistically superior courier or causal service-quality changes.

16 / RETURNS INTELLIGENCE

Return Requests by Recorded Reason

Distribution of recorded reasons across 1,200 return requests at the order level.



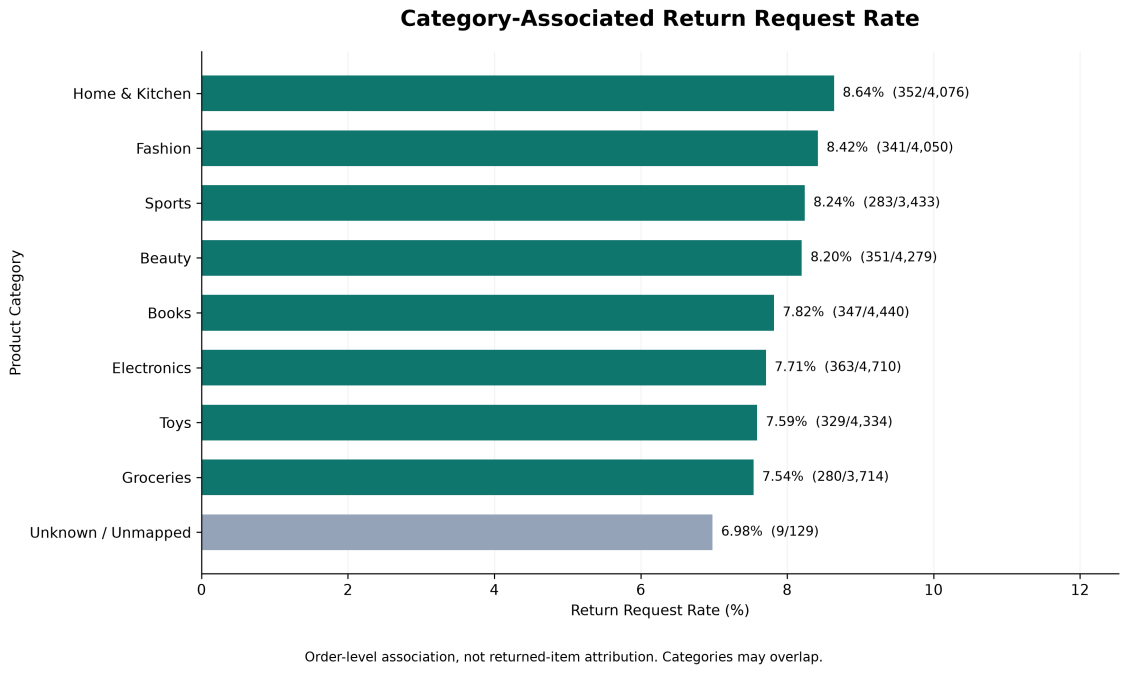
Business interpretation: Changed mind leads with 182 requests (15.17%), but no single reason represents most requests.

Reporting limit: These are recorded labels, not independently confirmed causes; requesting a return does not necessarily mean the return was completed.

17 / RETURNS INTELLIGENCE

Category-Associated Return Requests

Return-request associations grouped by categories included in the original order.



Business interpretation: Home & Kitchen: 352 of 4,076 orders containing the category also have a recorded return request = 8.64%.

Reporting limit: The request identifies the order, not the item returned. Orders with multiple product categories can be counted in multiple associations.

Read correctly: This is an order-category association percentage, not the product-level Home & Kitchen return rate.

18 / KEY TAKEAWAYS

Seven Key Takeaways

A concise management reading of the five charts and validated source-table findings.

01 Shipment visibility

Delivered is the dominant recorded shipment status at 77.87%.

02 Eligibility discipline

Only 11,122 shipments enter the baseline usable delivery-duration sample.

03 Chronology sensitivity

Screening reduces the comparison sample to 5,667 and shifts the courier ordering.

04 Return-request prevalence

8.00% of orders have a request; completed-return incidence is not measured.

05 Reason distribution

Changed mind is the leading recorded reason but only 15.17% of requests.

06 Category attribution

The 8.64% Home & Kitchen figure is an order-level association, not an item-level return rate.

07 Data reliability

Cross-system timestamp and status exceptions warrant reconciliation before confident service benchmarking.

19 / BUSINESS INSIGHTS

Evidence-Based Business Insights

These are the seven evidence-based findings recorded in the Python analytical export. Their limits are explicitly preserved rather than hidden in design.

I01 Shipment Status
11,680 of 15,000 shipments (77.87%) are marked Delivered.

I02 Delivery Lead Time
Average recorded delivery lead time is 4.00 days, with 11,122 eligible observations.

I03 Courier Sensitivity
The lowest average changes from NinjaVan to LBC after chronology screening; the screened courier mean range is 0.098 days.

I04 Return Request Rate
1,200 of 15,000 orders have a recorded return request (8.00%).

I05 Return Reasons
Changed mind is the most frequent reason, with 182 requests (15.17% of return records).

I06 Category Return Association
Home & Kitchen has the highest recorded order-category return-request association rate at 8.64%.

I07 Source Data Reliability
7,397 shipments precede their orders, 7,436 payments precede orders, and 938 cancelled orders have Delivered shipment status.

20 / RECOMMENDATIONS

Recommended Operational Priorities

The two proposed priorities address the strongest limitations. These are recommendations, not work already completed or measured financial benefits.

PRIORITY 01

Automated chronology and cross-system status reconciliation

Suggested accountable functions: Operations + Data Steward

Action: Validate linked order, payment, shipment and delivery events; flag cases before operational reporting.

Success measure: Number and percentage of unresolved chronology/status exceptions.

PRIORITY 02

Stronger returned-item identification and reason tracking

Suggested accountable functions: Returns + Customer Service

Action: Record returned product IDs and validate reasons; investigate persistent category patterns after item-level attribution.

Success measure: Return-ID completeness, reason completeness and validated item-level return rates.

Recommended sequence: Establish trustworthy source timelines first; then validate returned-item attribution and use category/courier comparisons for intervention design.

21 / CONCLUSIONS

Summary & Conclusions

SUMMARY

The Python workflow converted seven original CSV files into prepared datasets, documented quality flags, 13 executive KPIs, five charts, and an interactive dashboard. It supports descriptive visibility into shipment outcomes, delivery durations, courier variation, and recorded return requests.

CONCLUSIONS

Operational visibility is established

The baseline records answer what shipment states and return requests are currently recorded.

Courier ranking needs a stronger denominator

Different eligibility screens change the observed ordering; no definitive provider superiority is proved.

Return category insights remain associative

Order-level requests cannot identify which item or category was actually returned.

Trust controls precede performance action

Reconcile event times and inconsistent status combinations before managerial performance comparisons.

Decision statement: Strengthen the reliability of core operational event data and returned-item documentation; use current descriptive findings as diagnostic evidence, not verified intervention impact.

22 / METHODOLOGY APPENDIX

Metrics, Cohorts & Reporting Rules

Every measure below has a defined numerator, denominator, or eligible sample. This prevents a visitor from misreading one measure as another.

Metric	Population / exact definition	Verified sample
Delivered share	Delivered status records / all prepared shipment rows	11,680 / 15,000 = 77.87%
In-Transit / Lost shares	Relevant recorded status / all prepared shipment rows	17.96% / 1.07%
Lead time	Delivered date minus ship date, in days	11,122 eligible; mean 4.0035 days
Screened lead time	Baseline subset with order/shipping chronology check	5,667 eligible; mean 4.0194 days
Return request rate	Orders with request / distinct orders	1,200 / 15,000 = 8.00%
Return reason share	Requests assigned reason / all return requests	Changed mind 182 / 1,200 = 15.17%
Category association	Orders containing category with request / orders containing category	Home & Kitchen 352 / 4,076 = 8.64%
Courier average	Recorded lead-time mean among courier-eligible records	Descriptive metric; no significance claim

Filter behavior: Courier selection affects shipment-level and courier analyses. Whole-dataset returns and data-quality registers remain based on all relevant source records.

23 / TECHNICAL EVIDENCE

Selected Python Notebook Evidence

Selected original Python notebook code excerpts illustrate the cleaning rule and the final export safeguards. The full .ipynb is the reproducible source, rather than these abbreviated excerpts.

EXHIBIT A / DEDUPLICATION

```
# NOTEBOOK CELL 013 — CREATE DEDUPLICATED WORKING COPIES
customers_clean = customers.drop_duplicates().copy()
orders_clean = orders.drop_duplicates().copy()
order_items_clean = order_items.drop_duplicates().copy()
# All-column exact-match definition; originals remain unchanged.
```

EXHIBIT B / EXPORT VERIFICATION

```
# STEP 107 — FINAL CLEANED-DATA EXPORTS
file_path = cleaned_folder / filename
dataframe.to_csv(file_path, index=False)
readback = pd.read_csv(file_path)
assert len(readback) == len(dataframe)
assert list(readback.columns) == list(dataframe.columns)
```

Audit trace: Cleaned CSVs, the six-rule register, source-data quality exports, five PNG charts, and the STEP 109 validation CSV are included in the supplied project package.

24 / TECHNICAL EVIDENCE

Validation & Reproducibility Evidence

The latest root-level Logistics Analysis notebook in the uploaded ZIP contains STEP 107, STEP 108 and STEP 109. Its saved execution output confirms successful reconciliation.

ACTUAL SAVED JUPYTER OUTPUT / STEP 109

STEP 109 — FINAL VALIDATION PASSED

Cleaned datasets verified: 7 / 7

Dashboard exports verified: 4 / 4

Executive KPI checks: 13 / 13

Chart data and PNG files verified: 5 / 5

Cleaning evidence files verified: 3 / 3

Total validation checks passed: 85

CLEANED DATASETS	CLEANING RULES	RECONCILIATION CHECKS
7 / 7	6 / 6	85 / 85
STEP 107	STEP 108	STEP 109

Validation scope: Source-to-cleaned reconciliation, exported file hashes/structure, all 13 KPI values, five chart datasets and PNG integrity, four dashboard exports, and all three STEP 108 evidence files.

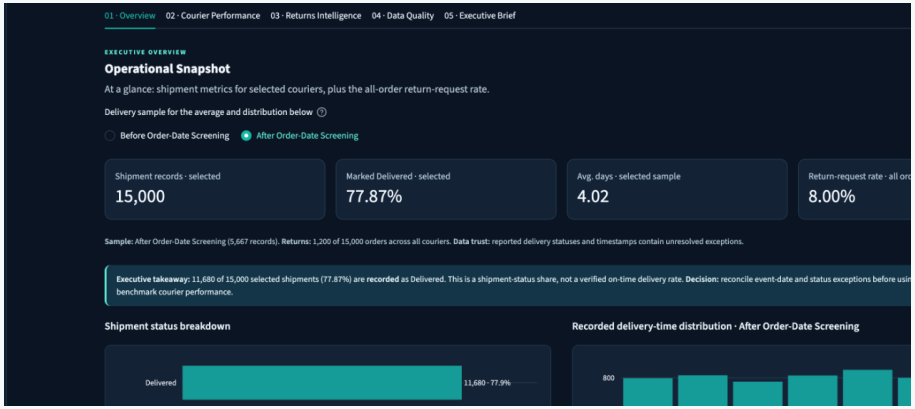
25 / DASHBOARD EVIDENCE

Six-Page Dashboard Evidence

Official application version: STEP 113. Five original analytical tabs were preserved and Page 06 - Data & Methodology was added. Six tabs are verified directly in the uploaded Streamlit source.

01 Overview | 02 Courier Performance | 03 Returns Intelligence
04 Data Quality | 05 Executive Brief | 06 Data & Methodology

APPLICATION EVIDENCE / ARCHIVED SCREEN + VERIFIED STEP 113 SOURCE



ARCHIVED OVERVIEW / captured analytics chart region

PAGE 06 / DATA & METHODOLOGY

01 Original source datasets

02 Original vs. prepared data

03 KPI definitions & calculation rules

04 Public datasets and evidence downloads

05 Python workflow and traceability

Evidence standard: The six-page navigation and Page 06 capabilities are verified in the uploaded STEP 113 Streamlit source; the visual excerpt is an archived overview capture. A refreshed clean full-window screenshot of the running six-page installation was not provided, so this report does not mislabel an old or recreated screen as a new live capture.